

# 具体患者未来轨迹 Demo: AF-FUTURE-001

## Demo

该 demo 使用预训练的Delphi模型，在一个具体测试集患者的基线时点隐藏其后续事件，并把模型在baseline事件后给出的 ranked token 与真实未来轨迹对照。

当前模型是序列 next-token 模型；这里展示的是基线时点的一步候选排序与后续真实轨迹的对应关系，不是固定 5 年绝对风险预测。

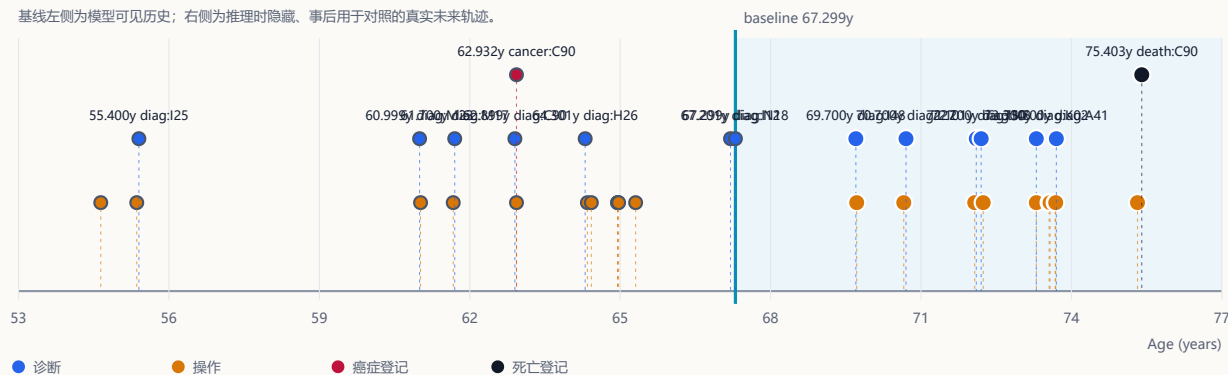
## 病例概览

| 项目                  | 内容                                           |
|---------------------|----------------------------------------------|
| 模型                  | Exp2                                         |
| 数据 split            | test                                         |
| split_patient_index | 2028                                         |
| 患者 ID               | 已脱敏，不在汇报材料中展示                                |
| 基线年龄                | 67.299 岁                                     |
| 首个隐藏真实事件            | diag:I48 at 69.700 岁                         |
| 目标展示疾病              | Atrial fibrillation and flutter, 房颤/房扑 (I48) |
| 首次房颤是否诊断 Top-5 命中   | 是                                            |
| baseline后随访长度       | 8.104 年                                      |
| baseline后死亡信息       | death:C90                                    |
| baseline后癌症信息       | baseline后未观察到癌症登记                            |

## 可视化

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基线左侧为模型可见历史；右侧为推理时隐藏、事后用于对照的真实未来轨迹。



## Baseline ranked candidates

Raw logits are only used for ranking within this model; green bars mean the token/group appears later in the hidden future.

### Top diagnosis candidates

|     |          |                                   |
|-----|----------|-----------------------------------|
| #1  | diag:N18 | logit -8.330; not observed later  |
| #2  | diag:N17 | logit -9.735; not observed later  |
| #3  | diag:I12 | logit -9.976; not observed later  |
| #4  | diag:E87 | logit -10.039; not observed later |
| #5  | diag:I48 | logit -10.287; observed later     |
| #6  | diag:I50 | logit -10.453; observed later     |
| #7  | diag:H26 | logit -10.690; not observed later |
| #8  | diag:I18 | logit -10.765; not observed later |
| #9  | diag:H25 | logit -10.795; not observed later |
| #10 | diag:D63 | logit -10.795; not observed later |

### Top selected disease groups

|     |                                             |                                   |
|-----|---------------------------------------------|-----------------------------------|
| #1  | Chronic kidney disease (N18)                | logit -8.330; not observed later  |
| #2  | Hypertension (I10-I15)                      | logit -9.976; not observed later  |
| #3  | Atrial fibrillation and flutter (I48)       | logit -10.287; observed later     |
| #4  | Heart failure (I50)                         | logit -10.453; observed later     |
| #5  | Ischemic heart disease (I20-I25)            | logit -11.223; not observed later |
| #6  | Chronic obstructive pulmonary disease (J44) | logit -11.828; not observed later |
| #7  | Type 2 diabetes (E11)                       | logit -11.836; not observed later |
| #8  | Myocardial infarction (I21-I22)             | logit -12.313; not observed later |
| #9  | Cerebrovascular disease (I60-I69)           | logit -13.280; not observed later |
| #10 | Disorders of lipoprotein metabolism (E78)   | logit -15.646; not observed later |

### Top death candidates

|    |           |                                   |
|----|-----------|-----------------------------------|
| #1 | death:C90 | logit -10.744; observed later     |
| #2 | death:C64 | logit -12.066; not observed later |
| #3 | death:I25 | logit -12.437; not observed later |
| #4 | death:I21 | logit -12.857; not observed later |
| #5 | death:C18 | logit -13.305; not observed later |
| #6 | death:E85 | logit -13.621; not observed later |
| #7 | death:I11 | logit -13.795; not observed later |
| #8 | death:C92 | logit -13.820; not observed later |
| #9 | death:C22 | logit -14.108; not observed later |

#10death:E14

logit -14.118; not observed later

Top cancer candidates



Actual future events scored at baseline

Ranks show where each true future token stood among same-type candidates at the baseline step.

| Age    | Actual future token | Type | Baseline logit | Rank within type |
|--------|---------------------|------|----------------|------------------|
| 69.700 | diag:I48            | 诊断   | -10.287        | 5                |
| 70.700 | diag:I27            | 诊断   | -12.770        | 83               |
| 72.101 | diag:I50            | 诊断   | -10.453        | 6                |
| 72.200 | diag:I48            | 诊断   | -10.287        | 5                |
| 73.300 | diag:K02            | 诊断   | -11.380        | 21               |
| 73.700 | diag:A41            | 诊断   | -10.827        | 11               |
| 75.403 | death:C90           | 死亡登记 | -10.744        | 1                |

Baseline前模型可见历史

| 年龄     | token      | 类型    |
|--------|------------|-------|
| 54.639 | proc:X998  | 手术/操作 |
| 55.357 | proc:K633  | 手术/操作 |
| 55.400 | diag:I25   | 诊断    |
| 60.999 | diag:M23   | 诊断    |
| 61.018 | proc:W822  | 手术/操作 |
| 61.670 | proc:W401  | 手术/操作 |
| 61.700 | diag:M17   | 诊断    |
| 62.899 | diag:C90   | 诊断    |
| 62.932 | cancer:C90 | 癌症登记  |
| 62.932 | proc:W365  | 手术/操作 |
| 64.301 | diag:H26   | 诊断    |
| 64.345 | proc:C751  | 手术/操作 |
| 64.424 | proc:K636  | 手术/操作 |
| 64.947 | proc:L916  | 手术/操作 |
| 64.950 | proc:X292  | 手术/操作 |
| 64.969 | proc:X352  | 手术/操作 |
| 65.309 | proc:X339  | 手术/操作 |
| 67.201 | diag:I12   | 诊断    |
| 67.299 | diag:N18   | 诊断    |

## Baseline后隐藏真实轨迹

| 年龄     | token     | 类型    |
|--------|-----------|-------|
| 69.700 | diag:I48  | 诊断    |
| 69.717 | proc:X501 | 手术/操作 |
| 70.656 | proc:K631 | 手术/操作 |
| 70.700 | diag:I27  | 诊断    |
| 72.071 | proc:U201 | 手术/操作 |
| 72.101 | diag:I50  | 诊断    |
| 72.200 | diag:I48  | 诊断    |
| 72.241 | proc:K623 | 手术/操作 |
| 73.300 | diag:K02  | 诊断    |
| 73.300 | proc:F168 | 手术/操作 |
| 73.555 | proc:X715 | 手术/操作 |
| 73.574 | proc:X724 | 手术/操作 |
| 73.676 | proc:X903 | 手术/操作 |
| 73.687 | proc:S132 | 手术/操作 |
| 73.700 | diag:A41  | 诊断    |
| 75.318 | proc:U051 | 手术/操作 |
| 75.403 | death:C90 | 死亡登记  |

模型输出：诊断 token

| Rank | token    | raw logit | 后续是否出现 | 首次出现年龄 |
|------|----------|-----------|--------|--------|
| 1    | diag:N18 | -8.330    | 否      |        |
| 2    | diag:N17 | -9.735    | 否      |        |
| 3    | diag:I12 | -9.976    | 否      |        |
| 4    | diag:E87 | -10.039   | 否      |        |
| 5    | diag:I48 | -10.287   | 是      | 69.700 |
| 6    | diag:I50 | -10.453   | 是      | 72.101 |
| 7    | diag:H26 | -10.690   | 否      |        |
| 8    | diag:J18 | -10.765   | 否      |        |
| 9    | diag:H25 | -10.795   | 否      |        |
| 10   | diag:D63 | -10.795   | 否      |        |

模型输出：有限疾病组

| Rank | 疾病组                                         | ICD-10  | 代表 token | raw logit | 诊断内排名 | 后续是否出现 | 首次出现               |
|------|---------------------------------------------|---------|----------|-----------|-------|--------|--------------------|
| 1    | Chronic kidney disease (N18)                | N18     | diag:N18 | -8.330    | 1     | 否      |                    |
| 2    | Hypertension (I10-I15)                      | I10-I15 | diag:I12 | -9.976    | 3     | 否      |                    |
| 3    | Atrial fibrillation and flutter (I48)       | I48     | diag:I48 | -10.287   | 5     | 是      | diag:I48<br>69.700 |
| 4    | Heart failure (I50)                         | I50     | diag:I50 | -10.453   | 6     | 是      | diag:I50<br>72.101 |
| 5    | Ischemic heart disease (I20-I25)            | I20-I25 | diag:I25 | -11.223   | 18    | 否      |                    |
| 6    | Chronic obstructive pulmonary disease (J44) | J44     | diag:J44 | -11.828   | 30    | 否      |                    |
| 7    | Type 2 diabetes (E11)                       | E11     | diag:E11 | -11.836   | 32    | 否      |                    |
| 8    | Myocardial infarction (I21-I22)             | I21-I22 | diag:I21 | -12.313   | 53    | 否      |                    |
| 9    | Cerebrovascular disease (I60-I69)           | I60-I69 | diag:I63 | -13.280   | 117   | 否      |                    |
| 10   | Disorders of lipoprotein metabolism (E78)   | E78     | diag:E78 | -15.646   | 345   | 否      |                    |

模型输出：死亡和癌症 token

| 类型 | Rank | token      | raw logit | 后续是否出现 | 首次出现年龄 |
|----|------|------------|-----------|--------|--------|
| 死亡 | 1    | death:C90  | -10.744   | 是      | 75.403 |
| 死亡 | 2    | death:C64  | -12.066   | 否      |        |
| 死亡 | 3    | death:I25  | -12.437   | 否      |        |
| 死亡 | 4    | death:I21  | -12.857   | 否      |        |
| 死亡 | 5    | death:C18  | -13.305   | 否      |        |
| 死亡 | 6    | death:E85  | -13.621   | 否      |        |
| 死亡 | 7    | death:I11  | -13.795   | 否      |        |
| 死亡 | 8    | death:C92  | -13.820   | 否      |        |
| 死亡 | 9    | death:C22  | -14.108   | 否      |        |
| 死亡 | 10   | death:E14  | -14.118   | 否      |        |
| 癌症 | 1    | cancer:C90 | -12.916   | 否      |        |
| 癌症 | 2    | cancer:C44 | -13.479   | 否      |        |
| 癌症 | 3    | cancer:D47 | -13.651   | 否      |        |
| 癌症 | 4    | cancer:D46 | -14.343   | 否      |        |
| 癌症 | 5    | cancer:C91 | -14.473   | 否      |        |
| 癌症 | 6    | cancer:C64 | -14.507   | 否      |        |
| 癌症 | 7    | cancer:D45 | -14.712   | 否      |        |
| 癌症 | 8    | cancer:D09 | -14.842   | 否      |        |
| 癌症 | 9    | cancer:C92 | -14.846   | 否      |        |
| 癌症 | 10   | cancer:C18 | -14.940   | 否      |        |

## 真实未来事件在baseline时间点时的得分

| 年龄     | 真实 token  | 类型   | baseline raw logit | 同类型候选内排名 |
|--------|-----------|------|--------------------|----------|
| 69.700 | diag:I48  | 诊断   | -10.287            | 5        |
| 70.700 | diag:I27  | 诊断   | -12.770            | 83       |
| 72.101 | diag:I50  | 诊断   | -10.453            | 6        |
| 72.200 | diag:I48  | 诊断   | -10.287            | 5        |
| 73.300 | diag:K02  | 诊断   | -11.380            | 21       |
| 73.700 | diag:A41  | 诊断   | -10.827            | 11       |
| 75.403 | death:C90 | 死亡登记 | -10.744            | 1        |

## 总结

在 67.299 岁baseline时间，模型只能看到此前病史；真实未来首个隐藏事件为 diag:I48 。模型对该目标 token 的诊断内排名为 5，因此这是一个 Top-5 命中的房颤展示病例。

该患者baseline之后真实轨迹还出现了 diag:A41 , diag:I27 , diag:I48 , diag:I50 , diag:K02 , 并最终记录 death:C90。

该 demo 可以展示具体患者层面的“模型候选排序”和“真实未来疾病/死亡轨迹”对照。